

DETAILED DESCRIPTION OF THE INVENTION — MASTER NEXALIS SYSTEM INTEGRATION AND MULTI-NODE ARCHITECTURE

The intraluminal longitudinal pelvic physiology monitoring device described herein may operate as a component within a broader multi-node physiologic intelligence system referred to collectively as the Nexalis System. The Nexalis System comprises multiple coordinated sensing nodes configured to acquire physiologic information from distinct anatomical regions and contextual environments.

In various embodiments, the disclosed device (Node 2) may communicate, synchronize, or otherwise cooperate with additional physiologic monitoring nodes including, but not limited to:

- Node 1 — Penile physiology monitoring systems configured to measure rigidity, tumescence, circulation, or related physiologic characteristics.
- Node 3 — Anatomical geometry or structural profiling systems configured to establish dimensional or functional reference profiles.
- Node 4 — Intraluminal compliance and distance measurement systems configured for internal spatial characterization.
- Node 5 — Upper-body or contextual physiologic monitoring nodes including chest-mounted, wearable, or environmental sensing systems.

The Nexalis System may integrate signals across nodes to generate composite physiologic models, compatibility analysis outputs, longitudinal health intelligence, or relational physiologic insights derived from synchronized multi-region monitoring.

Data generated by individual nodes may possess enhanced analytical value when interpreted collectively. Accordingly, the invention encompasses coordinated operation, shared computational analysis, distributed sensing architectures, and system-level intelligence derived from interactions between multiple nodes.

System coordination may occur through direct device-to-device communication, intermediary companion applications, cloud-based analytic platforms, edge computing environments, or hybrid distributed computational frameworks.

Embodiments described for one node may be applied to other nodes where technically feasible. Sensor technologies, communication systems, analytic methods, and operational workflows disclosed in connection with this device are intended to extend across the broader Nexalis System architecture.

The present disclosure therefore supports protection not only of individual device embodiments but also of integrated multi-node physiologic intelligence systems, coordinated monitoring platforms, and ecosystem-level implementations enabling unified physiologic analysis.

The Nexalis System integration concepts described herein are illustrative rather than limiting. Any arrangement of multiple physiologic sensing devices operating cooperatively to generate system-level physiologic intelligence is considered within the scope of the present disclosure.